Order of operations: BIDMAS

In this sheet you will…

- see why we need an agreed order for calculations
- use BIDMAS with brackets, indices and negative numbers
- work from left to right inside a pair
- write each step on its own line

Key words

order of operations
brackets
indices (powers)
left to right

This sheet follows our lesson step by step, so you can read it again at home.

1 The starter: who is right?

Maya and Leo both work out **2 + 3 × 4**. Maya gets 20. Leo gets 14.

Maya: left to right

2 + 3 = 5

5 × 4 = 20

20: wrong

Leo: × first

3 × 4 = 12

2 + 12 = 14

14: right

If everybody chose their own order, one calculation could have two answers. So we all agree on **one order**. Leo is right: 2 + 3 × 4 = 14.

2 The order: BIDMAS

We remember the order with the word **BIDMAS**. Always start at the top of the ladder.

B	Brackets	( )	do first
I	Indices squares and cubes	x^2 x^3	do next $x^2 = x \times x$ $x^3 = x \times x \times x$
D	Division	÷	do this pair next work from left to right
M	Multiplication	×	
A	Addition	+	do this pair last work from left to right
S	Subtraction	−	

Key idea × and ÷ share one step. + and − share one step. Inside a pair, work from left to right.

3 Indices with negative numbers

x^2 means $x \times x$ and x^3 means $x \times x \times x$. When the number is negative, **keep the brackets**.

Squared

$(-2)^2 = (-2) \times (-2) = 4$

Cubed

$(-2)^3 = (-2) \times (-2) \times (-2) = -8$

4 Putting it together

Worked examples: work out each calculation

a

$2 + 3 \times 4$
$= 2 + 12$
$= 14$

× comes before +, so work out 3 × 4 first. Then add.

b

$2 \times (3 + 4)$
$= 2 \times 7$
$= 14$

Brackets come first, so work out 3 + 4 first. Then multiply.

c

$24 \div 4 \times 3$
$= 6 \times 3$
$= 18$

÷ and × are a pair, so go from left to right: 24 ÷ 4 first, then × 3.

d

$-3 + 2 \times (-2)^2$
$= -3 + 2 \times 4$
$= -3 + 8$
$= 5$

Indices first: $(-2)^2 = 4$. Then ×. Then +. Keep the − in front of the first number.

Every line starts with "=" and has the same value as the line above it.

Watch out!

It is not just left to right. 2 + 3 × 4 is 14, not 20.

Pairs go left to right. 20 − 6 + 2 = 14 + 2 = 16. If you work out 6 + 2 first you get 12, which is wrong.

Keep the sign of a negative first number. −5 + 2 = −3.

One step per line. Johnny writes 5 + 7 = 12 − 10 = 2, but 5 + 7 is not equal to 2. Write 5 + 7 − 10 = 12 − 10 = 2.

Tip Underline the part you work out first. Then copy the rest of the line unchanged.

Summary checklist

- ☐ I know the order: brackets, indices, then × and ÷, then + and −.
- ☐ I work from left to right inside a pair.
- ☐ I keep the brackets when I square or cube a negative number.
- ☐ I write each step on its own line.

Practice Textbook exercises on the order of operations